

Ex 1.2.1.

Step 1: Goat, cabbage, ~~go~~ wolf
 Step 2: cabbage, wolf $\xrightarrow{\text{Goat}}$
 Step 3: cabbage $\xrightarrow{\text{wolf}}$ Goat
 Step 4: cabbage, $\xleftarrow{\text{Goat}}$ wolf
 Step 5: Goat $\xrightarrow{\text{cabbage}}$ wolf, cabbage
 Step 6: Goat $\xleftarrow{\quad}$ wolf, cabbage
 Step 7: $\xrightarrow{\text{Goat}}$ wolf, cabbage

Ex 1.2.2.

A, B, C, D takes 1, 2, 5, 10 minutes.

A, B, C, D min

A, B, C, D	$\xrightarrow{A, B}$	2.
C, D	$\xleftarrow{A}$	1
A	$\xrightarrow{C, D}$	10
A	$\xleftarrow{B}$	2
	$\xrightarrow{A, B}$	2

Ex 1.2.3.

Only option A because b and c require the angle of A angle. and c require ha is the height to base a.

Ex 1.2.6.

START.

Step 1: User initiates the call
 DISPLAY "Enter the phone number"
 INPUT phone-number

Step 2: Validate the phone number.
 IF phone-number is empty or invalid THEN
 DISPLAY "Invalid phone number"
 EXIT
 END IF



Step 3: Check network connectivity.

IF network is not available THEN

 DISPLAY "No network connection. Check your signal"

 EXIT

END IF

Step 4: Dial the number

 DISPLAY "Connecting..." "Dialing" + phone number

 WAIT FOR network response

 SEND phone-number to network

Step 5: WHILE call = wait for the network establish the call

 WHILE call-status is "Connecting"

 DISPLAY "connecting..."

 WAIT for network response

 END WHILE

Step 6: Check if call is connected

 IF call-status is "connected" THEN

 DISPLAY "Call connected"

 START call-timer

 WHILE call-status is "connected"

 Monitor call activity

 IF user presses "End Call" THEN

 SET call-status = "Ended"

 END IF

 END WHILE

 ELSEIF call-status is "Busy" THEN

 DISPLAY "The line is busy. Try later"

 ELSE IF call-status is "unreachable" THEN

 DISPLAY "The number is unreachable"

 ELSE

 DISPLAY "Call failed. Try later"

 END IF


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Step 7: If end call status is "Ended" THEN
    STOP call_timer
    DISPLAY "Call ended. Duration: call_timer"
    SEND "End call" signal to network.
ENDIF

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END

Ex 1.2.5.

- a) Step by step explanation.
1. Write down the binary number: Each digit in a binary number represents a power of 2, starting from 2^0 on the right.
 2. Assign powers of 2 to each digit.
 - The right most digit is multiply by 2^0 .
 - The next digit is multiplied by 2^1 and so on.
 3. Multiply each binary digit by its corresponding power by 2.
 - If the digit is 1, it contribute the value.
 - If the digit is 0, it contribute nothing.
 4. Sum all the value.

b) Pseudo code

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define sum = 0.
define index = head of string
origin = list(string).reverse
while listorigin[index].next != null:
    sum = origin[index] * pow(2, i)
    index += 1
end while
return sum.

```

Bài 3:

1. Multiplication: $a \times b$

Sử dụng phép để quy nguyên thủy.

$$\text{mult}(a, 0) = 0$$

$$\text{mult}(a, b+1) = \text{mult}(a, b) + a.$$



Hàm mult dc xây dựng từ phép cộng và phép hợp

2. Exponentiation: a^b

Sử dụng phép đệ quy

$$\text{exp}(a, 0) = 1.$$

$$\text{exp}(a, b+1) = \text{exp}(a, b) \times a.$$

Hàm exp dc xây dựng từ phép nhân và phép hợp

3. Factorial:

Sử dụng phép đệ quy nguyên thủy

$$\text{fact}(0) = 1$$

$$\text{fact}(a+1) = \text{fact}(a) \times (a+1)$$

4. Predecessor:

$$\text{pred}(0) = 0$$

$$\text{Định nghĩa: } \text{pred}(a) = \begin{cases} a-1 & \text{nếu } a > 0 \\ 0 & \text{nếu } a = 0 \end{cases}$$

$$\text{c/m: } \text{pred}(0) = 0.$$

$$\text{pred}(a+1) = a.$$

5. Proper subtraction: $a \perp b$

Sử dụng phép đệ quy nguyên thủy và hàm tiền nhân

$$a \perp 0 = a.$$

$$a \perp (b+1) = \text{pred}(a \perp b).$$

6. Minimum:

Sử dụng phép hợp và phép đệ quy nguyên thủy

$$\min(a, b) = a \perp (a \perp b)$$

$$\min(a_1, a_2, \dots, a_n) = \min(a_1, \min(a_2, \dots, a_n)).$$

Hàm min dc xây dựng từ phép trừ hợp lý và phép hợp

7. Maximum

Sử dụng phép hợp và phép đệ quy nguyên thủy

$$\max(a, b) = (a+b) - \min(a, b)$$

$$\max(a_1, \dots, a_n) = \max(a_1, \max(a_2, \dots, a_n)).$$

Hàm max dc xây dựng qua phép cộng, phép trừ hợp lý và phép hợp